

Solutions to JEE Advanced – 1 | JEE 2022 | Paper - 2

PHYSICS

Single digit integer (6 Questions)

1.(1) Taking x and y as shown, for motion of ball :

$$u_x = u \cos \theta + 5, u_y = u \sin \theta$$

$$a_x = -g \sin 37^\circ = -6$$

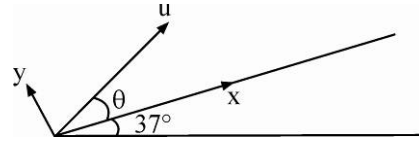
$$a_y = -g \cos 37^\circ = -8$$

At the centre of hoop, $S_y = 4$ and $V_y = 0$

$$\therefore V_y^2 = u_y^2 + 2a_y s_y \Rightarrow 0 = (u \sin \theta)^2 + 2(-8)(4)$$

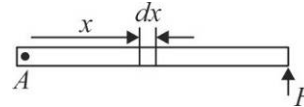
$$\Rightarrow u \sin \theta = 8$$

$$\text{Also } V_y = u_y + a_y t \Rightarrow 0 = u \sin \theta + (-8)t \Rightarrow t = \frac{u \sin \theta}{8} = \frac{8}{8} = 1$$



2.(3) When hinged at A

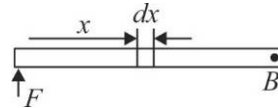
$$I_A = \int (dm)x^2 = \int_0^\ell \lambda_0 x dx x^2 = \frac{\lambda_0 \ell^4}{4}$$



$$\tau_A = I_A \alpha_A \Rightarrow \alpha_A = \frac{F \ell}{\left(\frac{\lambda_0 \ell^4}{4}\right)} = \frac{4F}{\lambda_0 \ell^3}$$

When hinged at B

$$I_B = \int (dm)(\ell - x)^2$$



$$= \int_0^\ell \lambda_0 x dx (\ell^2 + x^2 - 2\ell x) = \lambda_0 \left[\ell^2 \frac{\ell^2}{2} + \frac{\ell^4}{4} - 2\ell \frac{\ell^3}{3} \right] = \frac{\lambda_0 \ell^4}{12}$$

$$\tau_B = I_B \alpha_B \Rightarrow \alpha_B = \frac{F \ell}{\left(\frac{\lambda_0 \ell^4}{12}\right)} = \frac{12F}{\lambda_0 \ell^3} \quad \therefore \frac{\alpha_B}{\alpha_A} = \frac{12}{4} = 3$$

3.(3) For geostationary satellite: $r_1 = R + 6R = 7R$ $T_1 = 24h$

For second satellite: $r_2 = R + 0.75R = 1.75R$

Time period of satellite: $T = 2\pi \sqrt{\frac{r^3}{GM}}$ where, r is the radius of orbit

$$\Rightarrow \frac{T_2}{T_1} = \sqrt{\frac{r_2^3}{r_1^3}}$$

$$\frac{T_2}{24} = \sqrt{\left(\frac{1.75R}{7R}\right)^3} = \sqrt{\frac{1}{64}} = \frac{1}{8}$$

$$\therefore T_2 = \frac{24}{8} = 3 \text{ Hrs}$$

$$4.(1) \quad \rho g h A = m g ; h = \frac{m}{\rho A} = \frac{10}{1000 \times 100 \times 10^{-4}} = 1 \text{ m}$$

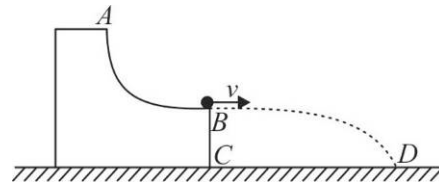
5.(8) Velocity with which it leaves the track can be found by energy conservation.

$$\frac{1}{2} m v^2 = m g R \Rightarrow v = \sqrt{2 g R}$$

After that, it undergoes a projectile motion till it hits the ground.

Time of flight will be

$$T = \sqrt{\frac{2h}{g}} ; \quad CD = \text{range} = vT = 2\sqrt{Rh} = 8 \text{ m}$$



6.(8) Work done = Change in kinetic energy

$$4(8) = \frac{1}{2} (1) v^2 \Rightarrow v = 8 \text{ m/s}$$

Multiple Correct (6 Questions)

$$7.(BC) \quad \frac{x}{v+u} = t_1 \Rightarrow v+u = \frac{x}{t_1} ; \quad \frac{x}{v-u} = t_2 \Rightarrow v-u = \frac{x}{t_2}$$

$$v = \frac{x}{2} \left(\frac{1}{t_1} + \frac{1}{t_2} \right) ; \quad u = \frac{x}{2} \left(\frac{1}{t_1} - \frac{1}{t_2} \right)$$

$$8.(ABC) \quad h = \frac{V_0}{A} = \frac{40}{8} = 5 \text{ cm}$$

Let speed of piston = u

Continuity equation

$$A u = a u_0 \Rightarrow u = \frac{a}{A} u_0 = \frac{2 \text{ mm}^2}{8 \text{ cm}^2} \times 2 \text{ m/s} = 0.005 \text{ m/s}$$

Applying Bernoulli's equation between points 1 and 2-

$$P_0 + \frac{1}{2} \rho u_0^2 + \rho g h = P + \frac{1}{2} \rho u^2$$

Where P is pressure above the piston at point 2.

$$\Rightarrow P = P_0 + \frac{1}{2} \rho u_0^2 - \frac{1}{2} \rho u^2 + \rho g h$$

For piston to go up at constant speed

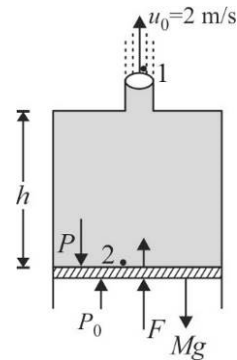
$$P_0 A + F = P A + M g \Rightarrow F = \frac{1}{2} \rho A (u_0^2 - u^2) + A \rho g h + M g$$

As piston moves up, h changes from 5 cm to zero. So, F keeps decreasing

$\therefore$ Work done by external agent is

$$W = - \int_h^0 F dh = \frac{1}{2} \rho A (u_0^2 - u^2) h + \frac{1}{2} A \rho g h^2 + M g h$$

$$= \frac{1}{2} \rho A u_0^2 h + \frac{1}{2} A \rho g h^2 + M g h \quad [\because u_0^2 \gg u^2] = 0.115 \text{ J}$$



9.(BD) Velocity of block is maximum when

$$mg = kx \Rightarrow x = \frac{10}{100} = 0.1 \text{ m}$$

Compression is maximum when velocity of block is zero.

Applying energy conservation

$$mg(h+x) = \frac{1}{2}kx^2$$

$$\Rightarrow 10(0.4+x) = 50x^2$$

$$\Rightarrow 5x^2 - x - 0.4 = 0$$

$$x = \frac{1 \pm \sqrt{1+8}}{10} = \frac{1+3}{10} = 0.4 \text{ m}$$

$$10.(BC) a = \frac{g \sin \theta}{1 + \frac{k^2}{R^2}}, t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{2h}{a \sin \theta}} \quad (\because s \sin \theta = h)$$

$$\text{For A, } \frac{2}{3}mR^2 = mk^2 \Rightarrow \frac{k^2}{R^2} = \frac{2}{3}$$

$$\therefore a_A = \frac{g \sin 60^\circ}{1 + \frac{2}{3}} = \frac{3\sqrt{3}g}{10}$$

$$t_A = \sqrt{\frac{2h}{\frac{3\sqrt{3}g}{10}}} \times \frac{\sqrt{3}}{2} = \sqrt{\frac{40h}{9g}}$$

$$\text{For B, } \frac{2}{5}mR^2 = mk^2 \Rightarrow \frac{k^2}{R^2} = \frac{2}{5}$$

$$\therefore a_B = \frac{g \sin 30^\circ}{1 + \frac{2}{5}} = \frac{5g}{14}$$

$$t_B = \sqrt{\frac{2h}{\frac{5g}{14} \times \frac{1}{2}}} = \sqrt{\frac{56h}{5g}} \quad \therefore t_B > t_A$$

$$k_A = k_B = mgh$$

$$k_A = \frac{1}{2}mv_A^2 + \frac{1}{2} \frac{2}{3}mR^2 \left(\frac{v_A}{R} \right)^2 = \frac{5}{6}mv_A^2$$

$$k_B = \frac{1}{2}mv_B^2 + \frac{1}{2} \frac{2}{5}mR^2 \left(\frac{v_B}{R} \right)^2 = \frac{7}{10}mv_B^2$$

$$\therefore \frac{5}{6}mv_A^2 = \frac{7}{10}mv_B^2 \Rightarrow v_B = \sqrt{\frac{25}{21}}v_A$$

11.(D) Let the volume and density of block be v & ρ_s and density of liquid be ρ_ℓ

On Earth : $w_1 = v\rho_s g$, $w_2 = v\rho_s g - v\rho_\ell g = v(\rho_s - \rho_\ell)g$, $w_1 - w_2 = v\rho_\ell g$, $w_1 / w_2 = \rho_s / (\rho_s - \rho_\ell)$

On moon : $w_1 = v\rho_s g / 6$, $w_2 = v\rho_s g / 6 - v\rho_\ell g / 6 = v(\rho_s - \rho_\ell)g / 6$,

$$w_1 - w_2 = v\rho_\ell g / 6, w_1 / w_2 = \rho_s / (\rho_s - \rho_\ell)$$

12.(BCD) (AB) $a = g \sin \theta - \mu g \cos \theta$

$$= 10 \times \frac{3}{5} - 0.5 \times 10 \times \frac{4}{5} = 2 \text{ m/s}^2$$

(C) $F_{\min} = mg \sin \theta - \mu mg \cos \theta = 10 \text{ N}$

(D) $mg \sin \theta = \mu(mg \cos \theta + F)$

$$30 = 0.5(40 + F) \Rightarrow F = 20 \text{ N}$$

13.(1.75) We know that $v_{\text{MAX}}^2 = 2(a)(x) \Rightarrow v_{\text{MAX}} = \sqrt{2ax}$

Time taken during the first segment, $T_1 = \frac{v_{\text{MAX}}}{a} = \sqrt{\frac{2x}{a}}$

Time taken during the second segment, $T_2 = \frac{x}{v_{\text{MAX}}} = \sqrt{\frac{x}{2a}}$

Time taken during the third segment, $T_3 = \frac{v_{\text{MAX}}}{\left(\frac{a}{2}\right)} = \sqrt{\frac{8x}{a}}$

Total time, $T = T_1 + T_2 + T_3 = \left(\sqrt{2} + \frac{1}{\sqrt{2}} + 2\sqrt{2}\right)\sqrt{\frac{x}{a}} = \frac{7}{\sqrt{2}}\sqrt{\frac{x}{a}}$

Distance travelled during the third segment, $X_3 = \frac{v_{\text{MAX}}^2}{2\left(\frac{a}{2}\right)} = 2x$

Total distance, $X = x + x + 2x = 4x$

Therefore, $v_{\text{AVG}} = \frac{\text{Total distance}}{\text{Total time}} = \frac{X}{T} = \frac{4}{7}\sqrt{2ax}$

So, $\frac{v_{\text{MAX}}}{v_{\text{AVG}}} = \frac{7}{4} = 1.75$

14.(0.60) For solid sphere

$$mv = I$$

$$Ih_1 = \frac{2}{5}mR^2\omega$$

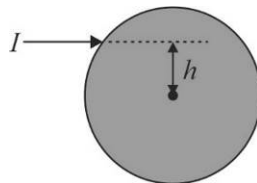
For $v = \omega R$,

$$\frac{I}{m} = \frac{5Ih_1}{2mR} \Rightarrow h_1 = \frac{2}{5}R$$

For hollow sphere

$$mv = I$$

$$Ih_2 = \frac{2}{3}mR^2\omega$$



For $v = \omega R$,

$$\frac{I}{m} = \frac{3Ih_2}{2mR} \Rightarrow h_2 = \frac{2}{3}R$$

$$\therefore \frac{h_1}{h_2} = \frac{3}{5} = 0.60$$

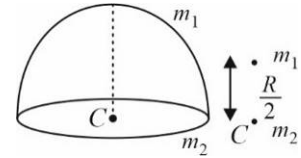
15.(0.33) Let σ be the mass per unit area (it will be same for the shell and the plate)

$$m_1 = \sigma(2\pi R^2), m_2 = \sigma(\pi R^2)$$

To find the CM, we can replace the hemispherical shell and the circular plate by point mass located at their CM as shown.

CM of the system from C

$$= \frac{m_1(R/2) + m_2(0)}{m_1 + m_2} = \frac{\sigma(2\pi R^2)(R/2)}{\sigma 3\pi R^2} = \frac{R}{3} = 0.33R$$



16.(0.16) Starting from the string directly attached to M , let the tension in the strings be T_1 , T_2 and T_3

Balancing torque on the pulleys,

$$T_1 r_4 = T_2 r_3 \Rightarrow Mgr_4 = T_2 r_3 \quad \dots (i)$$

$$T_2 r_2 = T_3 r_1 \Rightarrow T_2 r_2 = (mg \sin \theta) r_1 \quad \dots (ii)$$

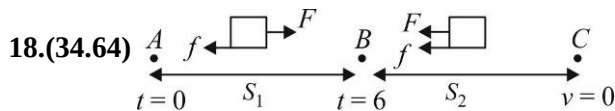
Multiplying the two equations, we get

$$(Mg)(r_4 r_2) = (mg \sin \theta)(r_3 r_1) \Rightarrow \frac{M}{m} = \frac{r_3 r_1}{r_4 r_2} \sin \theta = \frac{(4)(3)}{(10)(6)} \left(\frac{4}{5} \right) = 0.16$$

$$17.(3.46) y = x \tan \theta \left(1 - \frac{x}{R} \right)$$

Here $x = 3\text{m}$, $R = 3 + 6 = 9\text{m}$, $\tan \theta = \sqrt{3} = 1.73$

$$h = 2\sqrt{3} = 3.46\text{m}$$



From $A(t=0)$ to $B(t=6)$

$$a = \frac{F - f}{m} = 10^{-5} = 5 \text{ m/s}^2$$

$$s_1 = \frac{1}{2}at^2 = \frac{1}{2}(5)(36) = 90\text{m}$$

$$\text{Velocity at } B = at = 5(6) = 30 \text{ m/s}$$

After B , the direction of F is reversed.

As velocity is in forward direction, friction acts backwards till block comes to instantaneous rest at C .

From B to C

$$a = -\frac{(F + f)}{m} = -(10 + 5) = -15 \text{ m/s}^2$$

$$u = 30 \text{ m/s}$$

$$v^2 = u^2 + 2as_2$$

$$0 = (30)^2 + 2(-15)s_2 \Rightarrow s_2 = 30\text{m}$$

While returning from C to A, F acts towards left but friction acts towards right (as velocity is towards left)

$$s = -(s_1 + s_2) = -120\text{m}$$

$$a = -\frac{(F - f)}{m} = -5\text{m/s}^2$$

$$v^2 = u^2 + 2as = 0 + 2(-120)(-5)$$

$$v^2 = 1200 \Rightarrow v = 20\sqrt{3} = 20(1.732) = 34.64 \text{ m/s}$$

CHEMISTRY

- 1.(5) Polar compounds show deflection as per the figure because charged comb attracts polar molecules. Compounds having non-zero dipole moment are polar compounds. Polar compounds are: Trichloromethane (CHCl_3), Chlorobenzene ($\text{C}_6\text{H}_5\text{Cl}$), p-Dihydroxy benzene ($\text{p-HO-C}_6\text{H}_4\text{-OH}$), Benzenamine ($\text{C}_6\text{H}_5\text{NH}_2$) and Ethanol ($\text{C}_2\text{H}_5\text{OH}$).

Non-polar compounds are: Tetrachloromethane (CCl_4), Benzene (C_6H_6), p-Xylene ($\text{p-CH}_3\text{-C}_6\text{H}_4\text{-CH}_3$)

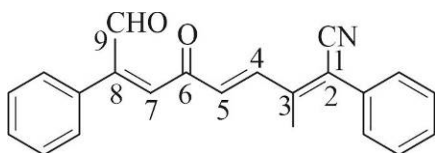
- 2.(6) $T = \text{constant}$

$$P_1V_1 = P_2V_2$$

$$7 \times 1 = \left(\frac{760}{760}\right) \times V_2 \Rightarrow V_2 = 7 \text{ litre}$$

$$W = -P(\Delta V) = -1 \times (7 - 1) = -6 \text{ Latm}$$

- 3.(0)

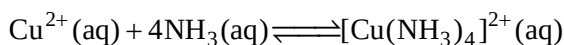


CN is principal functional group hence lowest locant will be given to C of CN.

$d = 2$, $b = 6$ and $a = 8$

- 4.(6) Electronic configuration of As, $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^3$. Electrons of 3p have $n = 3$ and $l = 1$.

- 5.(3) Reaction for formation of complex



$$K_f = \frac{[\text{Cu}(\text{NH}_3)_4]^{2+}}{[\text{Cu}^{2+}] \times [\text{NH}_3]^4} = \frac{(0.99)}{[\text{NH}_3]^4 \times 10^{-2}}$$

$$[\text{NH}_3]^4 = \frac{0.99}{1.1 \times 10^{11}}$$

$$[\text{NH}_3] = 0.001732 = 1.732 \times 10^{-3}$$

- 6.(3) The change in sign of radial wavefunction indicate the presence of nodes. Number of radial nodes = $(n - l - 1)$
For this orbital, number of radial nodes are 2 and it is ns orbital because $\Psi > 0$ at $r = 0$.

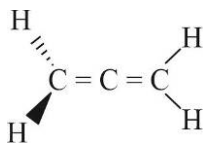
$$2 = (n - 0 - 1)$$

$$n = 3$$

7.(ACD) For mixing of two ideal gases at constant T and P, ΔU_{mix} and ΔH_{mix} are zero. During mixing, entropy of system increase while $q_{\text{mix}} = 0$.

8.(ACD) Mass independent properties are intensive properties.

9.(ABCD) Allene is nonplanar compound in which nodal plane of both the pi bonds are lying in different planes. Maximum three atoms are lying along the straight line because of non-planar structure. Hybridization state of all carbon atoms is not same. Terminal carbon atoms are sp^2 while middle carbon atom is sp hybridized.



10.(ABD) Symmetrical molecules have zero dipole moment. Structure of SF_4 is not symmetrical.

11.(BD) For homonuclear diatomic molecular species having 14 or a smaller number of electrons, bonding $\sigma(2p)$ orbital is higher in energy than $\pi(2p)$ orbitals.

12.(A) If $\Delta n_g = 0$ then there is no effect of change in pressure on equilibrium state.

13.(49) Z1 = 18, Z2 = 4, Z3 = 5, Z4 = 4, Z5 = 18

g sub shell is present in $n = 5$ principal energy level.

Minimum principal quantum number of an atomic orbital of g sub shell is 5. (Z3 = 5)

Azimuthal quantum number of an atomic orbital of g sub shell is 4. Because $l = 0$ to $n - 1$. (Z2 = 4)

Number of angular nodes in an atomic orbital of g sub shell are 4. Because number of angular nodes = l . (Z4 = 4)

Total 18 elements are expected in g block of the periodic table. (Z5 = 18)

Number of electrons in g sub shell are 18. Because number of electrons in a sub shell = $2(2l + 1)$. (Z1 = 18)

$$14.(0.38) K_c = \frac{[z]}{[x] \times [y]} = \frac{0.950}{1.20 \times 2.10} = 0.3769 = 0.38$$

$$15.(75) \text{ Mol of } CaCl_2 = \text{Mol of } CaCO_3 = \text{Mol of } CaO = \frac{0.56}{56} = 0.01$$

$$\text{Mass of } CaCl_2 = 0.01 \times 111 = 1.11$$

$$\text{Mass of NaCl} = 4.44 - 1.11 = 3.33$$

$$\% \text{ of NaCl} = \frac{3.33}{4.44} \times 100 = 75$$

$$16.(2.21) \text{ use } \lambda = \frac{h}{m \times v}, \text{ solve for } m$$

$$m = \frac{6.626}{3} \times 10^{-34} \times 10^{-8} \times 10^{10} = 2.20866 \times 10^{-32} \text{ kg} = 2.21 \times 10^{-32} \text{ kg}$$

$$17.(15.66) \text{ Calculated dipole moment} = 2.88 \times 10^{-29} \text{ C m}$$

$$\text{Observed dipole moment} = 1.35 \times 3.34 \times 10^{-30} \text{ C m} = 4.509 \times 10^{-30} \text{ C m}$$

$$\% \text{ ionic character} = \frac{4.509}{2.88} \times 10 = 15.65625 = 15.66$$

$$18.(3.44) \frac{P_1 \times V_1}{T_1} = \frac{P_2 \times V_2}{T_2} \quad V_2 = \frac{P_1 \times V_1 \times T_2}{P_2 \times T_1} = \frac{650 \times 5.50 \times 283}{980 \times 300} = 3.44$$

MATHEMATICS

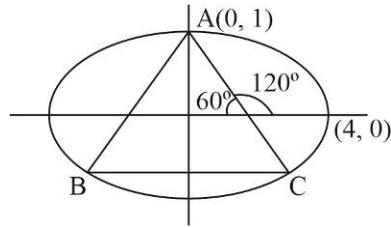
1.(8) Equation of AC is $\sqrt{3}x + y = 1$

$$\frac{x^2}{4} + (1 - \sqrt{3}x)^2 = 1$$

$$\frac{x^2}{4} + 3x^2 - 2\sqrt{3}x = 0$$

$$x = \frac{8\sqrt{3}}{13}, y = 1 - \frac{24}{13} = \frac{-11}{13}$$

$$AC = \frac{\sqrt{64 \times 3 + 576}}{13} = \frac{16\sqrt{3}}{13}$$



2.(8) $\frac{z_1}{z_2}$ is purely real Also $\left| \frac{z_1}{z_2} \right| = 1$

$$\Rightarrow \frac{z_1}{z_2} = \pm 1 \quad \dots\dots\dots(1)$$

$z_1 z_2$ is purely imaginary

$$|z_1 z_2| = 1 \Rightarrow z_1 z_2 = \pm i$$

$$\Rightarrow z_1^2 = \pm i$$

$$\Rightarrow z_1 = \pm e^{\pm i\pi/4}$$

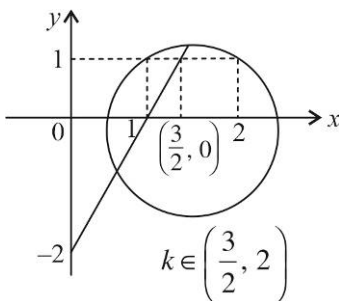
4 distinct possible values of z_1

And for each value of z_1 , z_2 can have 2 values $z_2 = \pm z_1$

$\Rightarrow$ 8 possible pairs of (z_1, z_2)

3.(0) $1 + \alpha^{10} + \alpha^{20} + \dots + \alpha^{190} = \frac{(\alpha^{10})^{20} - 1}{\alpha^{10} - 1} = \frac{(-1)^{20} - 1}{-1 - 1} = 0$

4.(3)



5.(8) $tx + y = at^3 + 2at$

Put $x = 3a$, $y = 0$ and we get $t = 0, \pm 1$

Also, put $x = 8$, $y = -2$ to get:

$$at^3 + 2(a - 4)t + 2 = 0$$

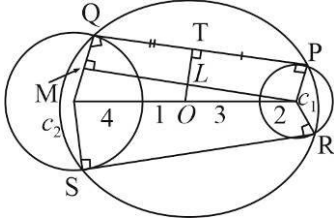
Only $t = 1$ will give us the correct value of $a = 2$

6.(1) $y^2 + 4y = x + 3$

$(y + 2)^2 = (x + 7)$

Vertex is at $(-7, -2) = (x_2, y_2)$ & focus is at $\left(-7 + \frac{1}{4}, -2\right) = \left(-\frac{27}{4}, -2\right) = (x_1, y_1)$

7.(AC)



$PQ = \sqrt{10^2 - (4 - 2)^2} = \sqrt{96} = 4\sqrt{6}$

T is mid point of PQ & TO parallel to C_1P & $C_2Q \Rightarrow O$ is mid point of C_1C_2

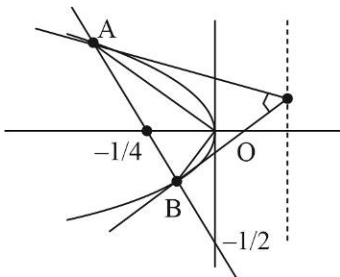
$OL = \sqrt{5^2 - (2\sqrt{6})^2} = 1$

$OT = 3$

$r = OP = \sqrt{3^2 + (2\sqrt{6})^2} = \sqrt{33}$

8.(CD) from $(h, 0)$ 3 distinct normals can be drawn to $y^2 = 4ax$ if $h > 2a$

9.(BC) Focus of $y^2 = -x$ is $\left(-\frac{1}{4}, 0\right)$ & hence $4x + 2y + 1 = 0$ is a focal chord



Homogenise $y^2 = -x$ using $4x + 2y + 1 = 0$

$y^2 = x(4x + 2y) \Rightarrow 4x^2 + 2xy - y^2 = 0$

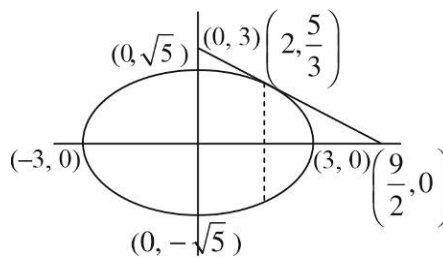
$|\tan \theta| = \frac{2\sqrt{1^2 - 4(-1)}}{4 - 1} = \frac{2\sqrt{5}}{3}$

10.(AD)

$A = 4 \times \frac{1}{2} \times \frac{9}{2} \times 3 = 27$

$C = 4 \times 2 \times \frac{5}{3} = \frac{40}{3}$

$D = 4 \times \frac{1}{2} \times 3 \times \sqrt{5} = 6\sqrt{5}$



$$\begin{aligned}
 11.(BCD) \quad |a + bw + cw^2|^2 &= (a + bw + cw^2)(a + bw^2 + cw) \\
 &= a^2 + b^2 + c^2 - ab - bc - ca \\
 &= \frac{1}{2}((a-b)^2 + (b-c)^2 + (c-a)^2)
 \end{aligned}$$

$$\begin{aligned}
 12.(ABC) \quad c^2 &= 16(1+m^2) = 25m^2 + 4 \\
 \Rightarrow 9m^2 &= 12 \Rightarrow m = \pm \frac{2\sqrt{3}}{3} \\
 c^2 &= 16\left(1 + \frac{12}{9}\right) = 16 \times \frac{7}{3} \\
 c &= \pm 4\sqrt{\frac{7}{3}} \\
 a = mc &= \frac{2}{\sqrt{3}} \times \frac{4\sqrt{7}}{\sqrt{3}} = \frac{8\sqrt{7}}{3}
 \end{aligned}$$

$$\begin{aligned}
 13.(44) \quad c_1(5, 12), r_1 &= \sqrt{25 + 144 - 120} = 7 \\
 c_2(0, 0), r_2 &= 20 \\
 c_1c_2 &= \sqrt{25 + 144} = 13 = r_2 - r_1 \\
 \Rightarrow \text{Circles touch each other internally} \\
 \Rightarrow 1 \text{ common tangent is given by } s - s' &= 0 \\
 \Rightarrow 5x + 12y = 260 \text{ (Directrix)} \\
 \Rightarrow \text{Distance of focus from directrix} = 2a &= \frac{+2 + 24 + 260}{13} = 22
 \end{aligned}$$

$$\begin{aligned}
 14.(43) \quad x^2 + y^2 - 6x - 6y + 14 &= 0 \quad x \geq 0, y \geq 0 \\
 x^2 + y^2 + 6x - 6y + 14 &= 0 \quad x \leq 0, y \geq 0 \\
 x^2 + y^2 - 6x + 6y + 14 &= 0 \quad x \geq 0, y \leq 0 \\
 x^2 + y^2 + 6x + 6y + 14 &= 0 \quad x \leq 0, y \leq 0
 \end{aligned}$$

$$\begin{aligned}
 \text{Let } x^2 + y^2 + 2gx + 2fy + c &= 0 \text{ intersect each of them orthogonally then} \\
 -6g - 6f &= c + 14 \\
 6g - 6f &= c + 14 \\
 -6g + 6f &= c + 14 \\
 6g + 6f &= c + 14 \\
 \Rightarrow c &= -14, g = f = 0 \\
 \text{Area} &= \pi \cdot 14 \approx 43.9
 \end{aligned}$$

$$\begin{aligned}
 15.(1.37) \quad z &= \frac{e^{\frac{i\pi}{12}} \times (-i) \cdot e^{\frac{i\pi}{3}}}{ie^{-\frac{i\pi}{4}} \times e^{-\frac{i\pi}{6}}} \\
 &= -e^{i\left(\frac{\pi}{12} + \frac{\pi}{3} + \frac{\pi}{4} + \frac{\pi}{6}\right)} = -e^{i\left(\frac{1+4+3+2}{12}\right)\pi} = -e^{\frac{5\pi}{6}} = -\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right) = -\left(-\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} - \frac{i}{2} \\
 \text{Re}(z) - \text{Im}(z) &= \frac{\sqrt{3} + 1}{2} \approx 1.37
 \end{aligned}$$

16.(3.50) Let $y = \tan 60^\circ x + c$ be the tangent

$$\text{Then } c^2 = 16 \cdot 3 + 1 = 49 \Rightarrow c = \pm 7$$

$$y = \sqrt{3}x + 7, y = \sqrt{3}x - 7$$

$$\text{Points of contact } A\left(-\frac{\sqrt{3} \cdot 16}{7}, \frac{1}{7}\right) \text{ or } B\left(\frac{\sqrt{3} \cdot 16}{7}, -\frac{1}{7}\right)$$

$$AB = \frac{2\sqrt{769}}{7}$$

17.(5.60) Consider family

$$(x-3)^2 + (y-5)^2 + \lambda(2x-y-1) = 0$$

$$x^2 + y^2 + 2(\lambda-3)x - (10+\lambda)y + 34 - \lambda = 0$$

$$\text{Its centre is } \left(3-\lambda, 5+\frac{\lambda}{2}\right), \text{ if it lies on } x+y-5=0, \text{ then } \lambda = 6$$

$$x^2 + y^2 + 6x - 16y + 28 = 0$$

18.(40) $\sqrt{5+12i} = \pm(3+2i)$

$$\sqrt{5-12i} = \pm(3-2i)$$

$$\Rightarrow \sqrt{5+12i} - \sqrt{5-12i} = 4i, +6, -6, -4i$$